

# UPLOTS: A Unified Pretrained Language Model for Constrained Time-series Generation

Anonymous Author(s)

## Abstract

In time series generation, existing approaches typically handcraft or train a separate model for each dataset, which hinders their scalability and fails to leverage shared temporal structures across domains. To address this fragmentation, we propose UPLOTS, a Unified, Prompt-guided Language model framework fOr constrained Time-Series Generation across diverse domains. Instead of building task-specific models, UPLOTS leverages a single pre-trained transformer backbone guided by learned constraint prompts, enabling on-demand generation with precise pattern control. One key innovation is our dynamic multi-dataset loss re-weighting and prompt-to-pattern mapping, which allows UPLOTS to internalize inverse temporal structures during training and conditionally generate them at inference. We evaluate UPLOTS on four real-world benchmarks (ETTh, Energy, PEMS04, and PEMS08) and two critical peak patterns, demonstrating superior fidelity, pattern compliance, and generalization against multiple SOTA baselines and single-dataset variants. The experiment results position UPLOTS as a flexible, scalable model for time series synthesis, demonstrating the potential of a single prompt-driven LLM to learn, adapt, and generate across domains.

Our code and baselines are available at anonymous github repo: <https://anonymous.4open.science/r/UPLOTS-6C36>.

## CCS Concepts

• Information systems → Sensor networks.

## Keywords

Time-series Generation, Language Model, Unified Model

## ACM Reference Format:

Anonymous Author(s). 2018. UPLOTS: A Unified Pretrained Language Model for Constrained Time-series Generation. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

## 1 Introduction

Time series generation plays a crucial role in diverse applications spanning finance, transportation modeling, and healthcare analytics. Existing methods [2] typically rely on designing or training a distinct model for each specific dataset or task, resulting in limited

scalability and inefficient exploitation of common temporal structures across various domains. This fragmentation limits the model's capability to assimilate extensive knowledge from varied datasets, thereby restricting its performance in time series generation tasks.

Recent VAE, GAN, and diffusion-based models have catalyzed notable progress in time series generation, expanding the repertoire of generative mechanisms and delivering richer, more flexible representations of complex temporal dynamics. Despite continued advances in VAE, GAN and diffusion-based approaches to time series modeling and generation, these methods remain limited in generalization and persist in the 'one model per dataset' paradigm illustrated on the left of Figure 1. As the number of diverse datasets grows, the need to train, tune, and deploy a separate model for each dataset leads to linear or even superlinear increases in computational and human effort, rendering large scale applications impractical. While a few recent studies have explored prompt driven or prototype driven transfer, they continue to rely on domain specific backbones or exemplar series, leaving open the quest for a single, text-only generator capable of constraint-aware synthesis across unseen domains.

Inspired by the success of prompt driven generation in NLP, we break free from purely diffusion or Transformer only architectures by adopting a single pretrained large language model (LLM) as a unified backbone. Dataset specific constraints are encoded as natural language prompts and fused with time series inputs via a prompt embedding module, allowing the model to be trained on a mixture of all temporal patterns and, at inference time, to generate customized target sequences on demand. This hybrid design combines the stability and fidelity of diffusion-style self-supervised objectives with the adaptability and efficiency of prompt-guided LLM inference, fundamentally addressing the instability of GANs, the mitigating earlier reports of over-smoothed outputs in some VAE variants, while recent models such as KoVAE [30] shows that well-designed VAEs can match or exceed diffusion models in fidelity. The high latency of pure diffusion pipelines, and delivering a scalable, unified solution for constrained time series generation.

To establish a robust unified framework, we propose UPLOTS illustrated on the right of Figure 1, which leverages a pretrained LLM to jointly learn from heterogeneous time series datasets, capturing the temporal patterns that respect each domain's distinct dynamics. To enable on-demand generation of customized target sequences at inference time without retraining, we encode dataset-specific constraints as lightweight adaptation of a pretrained transformer (train LN + positional embeddings). Using Dynamic Weighted Training Strategy that includes the Curriculum-Aware Loss Modulation and Rolling-Loss-Based Dynamic Sampling in multi-dataset learning helps to down-weight noisy and trivial tasks early in training that allocate more updates to underperforming datasets.

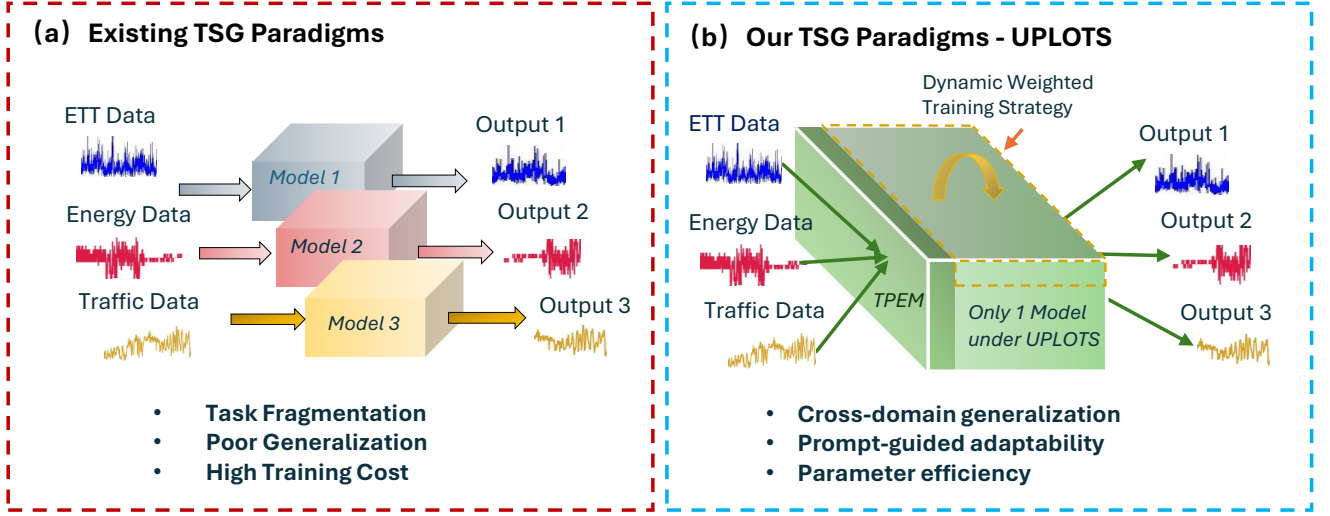
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](mailto:permissions@acm.org).

Conference acronym 'XX, Woodstock, NY

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>



**Figure 1: Conceptual comparison between traditional time series generation (TSG) methods and our proposed UPLOTS framework.** The left (red frame) illustrates the limitations of conventional approaches, including *task fragmentation*, *poor generalization* due to the need for dataset-specific models, where a dedicated model is typically required for each dataset. The right (blue frame) shows the improvements of UPLOTS: a *unified training pipeline*, *cross-domain generalization*, and *prompt-guided adaptability*. The integration of the Time-series Prompt Embedding Module (TPEM, green block) enables UPLOTS to flexibly condition on diverse temporal patterns via prompts.

## 1.1 Contribution

- We propose a unified LLM Framework based on Large Language Models (LLMs) that enables training across all datasets and inference on test sets with diverse objectives. This framework breaks the traditional one-to-one paradigm in unconditional time series generation models, where a dedicated model is typically required for each dataset.
- Building on unconditional generation, our framework leverages prompt learning and multi-dataset training to empower on-demand, task-specific generation. For example, the model can directly generate morning peak time series segments for the Energy dataset using textual prompts, achieving superior performance compared to single-dataset single-model approaches. Furthermore, the model can reuse diverse prompts to generate different target time series without retraining.
- Our framework demonstrates significant advantages over state-of-the-art (SOTA) baselines and its basic model variants across four real world datasets and two critical temporal patterns (morning peak and evening peak), showcasing robustness and generalizability.

## 2 Related work

### 2.1 Time-series Generation

**VAE.** C-RNN-VAE [9] fuses a VAE with conditional RNN decoders for coherent sequence synthesis, while VRNN [5] and SRNN [11] enrich the latent space with recurrent and hierarchical dynamics. Later variants mature the idea: GP-VAE [10] adopts Gaussian-process priors, TimeVAE [7] adds a dual-branch decoder, and Time-Causal VAE [1] imposes causality with a RealNVP prior; the latest

models—Koopman VAE [30], TimeAutoDiff [35], TimeVQVAE [23], and Time-Based VQVAE [28]—inject Koopman theory, diffusion, or vector-quantization mechanisms to further improve temporal fidelity. Although early VAEs often produced blurred sequences, recent advances such as KoVAE [30] and TimeVQVAE [23] employ Koopman dynamics or vector quantisation to deliver high-fidelity samples comparable with diffusion approaches.

**GAN.** In 2019, C-RNN-GAN [27] first validated adversarial training for continuous sequences, followed by LSTM-based RGAN/R-CGAN [8] for medical time series and TimeGAN [40]’s hybrid supervised-adversarial approach. Recent advances include GT-GAN [20] (global-local critics), PSA-GAN [19] (hierarchical self-attention), RTSGAN [32] (encoder-decoder for irregular series), and COT-GAN [38] (causal optimal transport). GANs show that adversarial training can produce realistic sequences, but they suffer from unstable training, mode collapse, and poor long-range coherence—factors that make them less reliable and increasingly unpopular in recent benchmarks.

**DDPM.** DDPM models generate sequences by iterative denoising: DiffWave [22] enforces monotonic or periodic structure, DiffTime [6] enables conditional forecasting, Diffusion-TS [41] adds a transformer with disentangled temporal cues, WaveStitch [34] accelerates sampling via segment stitching, D<sup>3</sup>M [39] fuses discrete diffusion with continuous flows, and DiffImp [13] targets imputation with a bidirectional Mamba core.

Furthermore, SDformer [4] discretizes series for fast Transformer decoding, ImagenTime [29] converts sequences to images to reuse vision diffusers. FIDE [12] combines the diffusion model with frequency-domain inflation and block-maxima conditioning with GEV integration to faithfully generate extreme events. TimeDP [18] conditions

generation through domain-prompt motifs—illustrating how NLP and CV ideas enrich time series synthesis. DDPMs deliver high-fidelity samples and enforce structural constraints via iterative denoising, yet their costly multi-step sampling and rigid conditioning limit real-time use and flexible control. T2S [15] proposes a diffusion-based Text-to-Series model with a length-adaptive VAE and Diffusion Transformer, trained on a new 600K+ dataset to generate arbitrary-length time series from text.

## 2.2 LLM for Time-series Models

LLMs now tackle time series by “reading” numbers as text. Gruver et al [17] showed that digit-stream tokenization lets GPT-3 [3] and LLaMA-2 [37] deliver zero-shot forecasts competitive with bespoke models. Prompt engineering generalizes this idea: GPT4TS [42] handles forecasting, classification, and anomaly detection; UniTime [26] unifies cross-domain tasks via template-guided fine-tuning; TEST [36] aligns numeric prototypes with LLM embeddings; and TimeLLM [21] reprograms frozen models through “Prompt-as-Prefix.” Scaling up, TimeGPT [14] pre-trains on approximate 100 B time points, and CEHR-GPT [31] generates privacy-preserving EHR sequences. PaD-TS [25] adds dual encoders and a population-preservation loss to diffusion models, while BRIDGE [24] couples multi-agent LLM prompt refinement with diffusion-based generation. Together, these works span prompt-only, hybrid, and fully pre-trained paradigms that extend LLMs to versatile, cross-domain time series synthesis and analysis.

These studies highlight that LLM has shown strong capabilities in time series forecasting and classification, but time series generation especially constrained generation is less studied. Our Unified Pretrained LM Framework fills this gap, enabling frozen LLM to generate time series that conform to the constrained target.

## 3 Preliminaries

### 3.1 Problem definition

A time series is represented as

$$X_{1:\tau} = (x_1, \dots, x_\tau) \in \mathbb{R}^{\tau \times d}, \quad (1)$$

where  $\tau$  denotes the number of timesteps and  $d$  denotes the feature dimension at each timestep. After splitting the raw data into  $n$  segments, we obtain a collection of time-series segments:  $D = \{X_{1:\tau}^i\}_{i=1}^n \in \mathbb{R}^{\tau \times d}$ .

In the training stage of the time-series generation task, we feed  $D$  into a model  $\mathcal{F}$  and optimize its parameters  $\theta$  so that  $\mathcal{F}$  can generate time series whose distribution matches that of  $D$ . At inference time, the original dataset  $D$  is no longer needed and we simply specify the desired output shape and use the learned model  $\mathcal{F}_\theta$  to perform unconditional generation of new time-series segments:

$$\hat{X}_{1:\tau} = \mathcal{F}_\theta(\text{shape} = (n, \tau, d)). \quad (2)$$

### 3.2 Basic TS Generation Model: Diffusion-TS

Diffusion-TS proceeds in three steps:

*Forward diffusion.* A clean sequence  $x_0 \in \mathbb{R}^{\tau \times d}$  is progressively noised via  $q(x_t | x_0) = \mathcal{N}(\sqrt{\alpha_t} x_0, (1 - \alpha_t) \mathbf{I})$ ,  $t = 1, \dots, \tau_d$  with a fixed variance schedule  $\{\alpha_t\}$ , where  $t$  is the diffusion step.

*Reverse model and loss.* A Transformer  $\mathcal{F}_\theta(x_t, t)$  predicts the original signal rather than the noise,  $\hat{x}_0 = \mathcal{F}_\theta(x_t, t)$ . Training minimises a joint time/frequency objective

$$\mathcal{L} = \underbrace{\|x_t - \hat{x}_t\|_2^2}_{\text{time domain}} + \lambda \underbrace{\|\mathcal{F}(x_t) - \mathcal{F}(\hat{x}_t)\|_2^2}_{\text{frequency domain}}, \quad (3)$$

which preserves pointwise accuracy and spectral content.

*Interpretable decoder.* The network’s output is factorised into trend, seasonality, and residual—so each generated sequence comes with an explicit trend/season breakdown:

$$\hat{x}_0 = \underbrace{P(k)}_{\text{trend}} + \underbrace{\sum_{m=1}^K A_m \cos(2\pi f_m k + \phi_m)}_{\text{seasonality}} + \underbrace{r(k)}_{\text{residual}} \quad (4)$$

## 4 Methodology

### 4.1 Unified Pretrained LM Framework

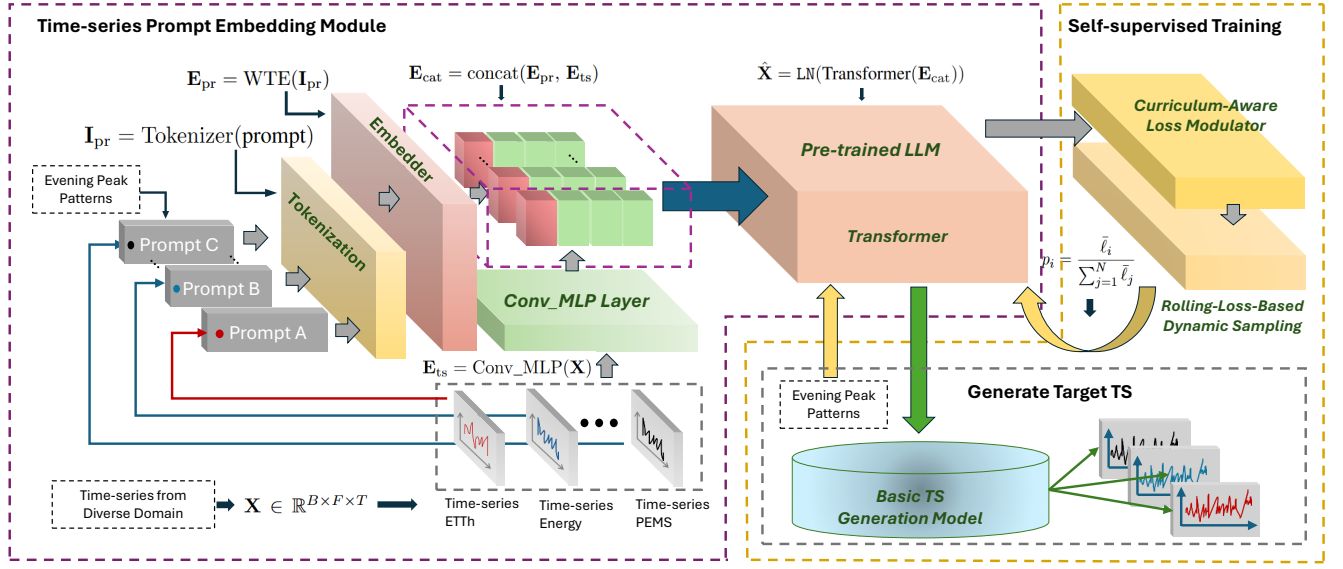
As shown by Figure 2, traditional time series generation (TSG) methods tend to train individual models on individual datasets separately, optimized for sequence generation specifically for the features of their respective dataset. Even though straightforward, separate training by definition limits models’ generalization capability and flexibility over a broad cross section of time series domains.

Our proposed framework offers a Unified Multi-dataset Representation, which constructs a universal embedding space, enabling effective generalization over disparate time series datasets. With Prompt-driven Learning, dataset-specific prompts are encoded into a shared representation, making it contextually adaptive so that it generates personalized sequences consistent with provided temporal patterns at inference. The framework also incorporates a Dynamic Weighted Training Strategy, which combines Curriculum-Aware Loss Modulation with Rolling-Loss-Based Dynamic Sampling, enabling effective convergence and well-balanced performance over disparate datasets. GPT-2 [33] and LLaMa3.2-1B [16] are used by us as the Pre-trained LLM. Overall, our Unified Pretrained LM Framework represents a significant leap forward in TSG, offering a scalable, adaptive, and highly generalizable solution for real-world applications. The framework demonstrates strong scalability and generalization across diverse time series datasets, thanks to its unified architecture and flexible training strategy. This ensures robust performance in dynamic, heterogeneous data environments, making it ideal for practical deployment.

### 4.2 Time-Series Prompt Embedding Module

In our unified framework, multiple diverse time series datasets are jointly trained within a single pre-trained LLM based model. Each dataset retains its unique temporal dynamics through natural language prompts. During inference, these prompts serve as domain-specific conditioning signals, empowering the model to flexibly generate context-aware sequences tailored to various target time series scenarios.

The proposed *Time-series Prompt Embedding Module* forms the core component of the entire framework. The module uses a deep



**Figure 2: The overview of UPLOTS comprises a data-transformation pipeline and its supporting techniques. During training, time series sequences and their domain-specific prompts are encoded by the Time-Series Prompt Embedding Module, yielding representations compatible with any baseline generator. Coupled with the Curriculum-Aware Loss Modulator and Rolling-Loss Dynamic Sampling, this design enables a single model to produce target sequences that obey the supplied prompts.**

feature fusion mechanism to organically combine prompts with high dimensional time series embedding representation processed by dimension projection, and in the process accurately establishes a bidirectional mapping relationship between time series dynamic features and semantic prompts. The shared model backbone exhibits dual modeling capabilities: it concurrently learns both domain-invariant temporal patterns and dataset-specific dynamics conditioned on their respective semantic prompts.

Given an input tensor  $X \in \mathbb{R}^{B \times F \times T}$ , where  $B$  is the batch size,  $F$  the feature dimension, and  $T$  the sequence length, we first project the sequence into the model’s hidden dimension  $D$  via a convolutional MLP:

$$E_{ts} = \text{Conv\_MLP}(X) \in \mathbb{R}^{B \times T \times D}. \quad (5)$$

A textual prompt (e.g., “Morning Peak Patterns in <DATASET>”) is tokenized using Tokenizer and embedded via LLM’s word embedding layer to convert the prompt into prompt embeddings:

$$I_{pr} = \text{Tokenizer}(\text{prompt}) \in \mathbb{Z}^{B \times L}, \quad (6)$$

where  $B$  is the batch size and  $L$  is the prompt length. These IDs are then passed through the frozen word-embedding table of the pretrained LLM,

$$E_{pr} = \text{WTE}(I_{pr}) \in \mathbb{R}^{B \times L \times D}, \quad (7)$$

yielding a dense prompt-embedding tensor  $E_{pr}$  that preserves the semantic cues required for subsequent fusion with the time-series embeddings.  $L$  denotes the tokenized prompt length. To improve efficiency, we retain only the first two transformer layers of GPT-2

or employ a lightweight 1B-parameter version of LLaMA for inference. All model parameters are frozen, except for the LayerNorm and positional embeddings.

We concatenate the prompt and sequence embeddings along the temporal dimension:

$$Z_0 = [E_p(P); E_x(X)] \in \mathbb{R}^{B \times (L+T) \times d_h}, \quad (8)$$

where  $[\cdot; \cdot]$  denotes concatenation along the temporal dimension.

The processed representations  $Z_0$  are then fed into a truncated pre-trained LLM’s Transformer architecture. Then the linear projection layers (LN) subsequently map the transformer outputs back to the original feature space:

$$\hat{X} = \text{LN}(\text{Transformer}(Z_0)) \in \mathbb{R}^{B \times (L+T) \times F}. \quad (9)$$

The output of Equation 9 will be the input of Diffusion-TS as mentioned in Section 3.1

With this module, any basic model can gain flexible cross-domain adaptability. The resulting high-dimensional features can be seamlessly integrated with a fundamental time series generation model for further feature extraction and modeling.

### 4.3 Dynamic Weighted Training Strategy

**4.3.1 Curriculum-Aware Loss Modulator.** For each dataset  $i$  we keep a deque  $\mathcal{H}_i$  of the most recent  $W$  losses and compute the running mean

$$\bar{\ell}_i = \begin{cases} \frac{1}{|\mathcal{H}_i|} \sum_{k \in \mathcal{H}_i} \ell_k, & |\mathcal{H}_i| > 0, \\ 1.0, & |\mathcal{H}_i| = 0, \end{cases} \quad (10)$$



**Table 1: Comparison with baselines. Context-FID, Discriminative-Score and Predictive-Score are displayed in both single data and multi data training mode. Lower values indicate better performance. Values in bold and underlined indicate the best performance.**

| Metric               | Data Type   | Model                 | ETTh_MP            | ETTh_EP            | Energy_MP          | Energy_EP          | PEMS04_MP          | PEMS04_EP          | PEMS08_MP          | PEMS08_EP          |
|----------------------|-------------|-----------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Context-FID          | Single Data | TimeGAN-NIPS2019      | 0.2939±.024        | 0.1365±.019        | 0.2831±.033        | 0.2512±.029        | 0.0959±.017        | 0.0619±.018        | 0.0741±.016        | 0.4383±.054        |
|                      |             | TimeVAE-Arxiv2022     | 0.6352±.074        | 0.2634±.046        | 0.3925±.036        | 0.4712±.044        | 0.4751±.051        | 0.7330±.084        | 0.3768±.026        | 0.4882±.033        |
|                      |             | Diffwave-ICLR2021     | 0.1613±.016        | 0.1081±.014        | 0.1125±.011        | 0.1354±.019        | 0.0874±.009        | 0.0995±.008        | 0.8951±.012        | 0.9541±.077        |
|                      |             | DiffTime-NIPS2023     | 0.1199±.009        | 0.0652±.008        | 0.0439±.003        | 0.0526±.007        | 0.0418±.008        | 0.0866±.013        | 0.0700±.011        | 0.0757±.009        |
|                      |             | Diffusion-TS-ICLR2024 | 0.0478±.005        | 0.0153±.002        | 0.0462±.007        | 0.0419±.003        | 0.0275±.005        | 0.0139±.003        | 0.0252±.004        | 0.5037±.081        |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.2145±.058        | 0.0951±.024        | 0.0840±.024        | 0.0856±.025        | 0.1271±.012        | 0.1417±.012        | 0.1176±.026        | 0.2340±.054        |
|                      |             | UPLOTS (Empty)        | 0.1682±.031        | 0.1997±.016        | 0.2908±.025        | 0.2040±.017        | 0.3285±.027        | 0.6286±.074        | 0.3276±.051        | 0.5256±.053        |
|                      |             | UPLOTS (GPT2)         | 0.0243±.002        | <b>0.0136±.001</b> | 0.0348±.004        | 0.0334±.002        | 0.0161±.002        | 0.0073±.001        | <b>0.0137±.001</b> | <b>0.0091±.000</b> |
|                      |             | UPLOTS (llama3)       | <b>0.0226±.002</b> | 0.0145±.002        | <b>0.0323±.002</b> | <b>0.0315±.002</b> | <b>0.0146±.002</b> | <b>0.0064±.001</b> | 0.0141±.001        | 0.0093±.000        |
|                      |             |                       |                    |                    |                    |                    |                    |                    |                    |                    |
| Discriminative Score | Single Data | TimeGAN-NIPS2019      | 0.0299±.016        | 0.0151±.009        | 0.0112±.009        | 0.0605±.007        | 0.0188±.016        | 0.0402±.024        | 0.0126±.002        | 0.0270±.024        |
|                      |             | TimeVAE-Arxiv2022     | 0.0232±.016        | 0.0135±.011        | 0.0223±.012        | 0.0303±.007        | 0.0188±.016        | 0.0716±.054        | 0.0154±.009        | 0.0453±.016        |
|                      |             | Diffwave-ICLR2021     | 0.0158±.010        | 0.0161±.013        | 0.0194±.009        | 0.0157±.011        | 0.0236±.021        | 0.0187±.014        | 0.0249±.006        | 0.0104±.009        |
|                      |             | DiffTime-NIPS2023     | 0.0119±.008        | 0.0140±.010        | 0.0172±.014        | 0.0175±.013        | 0.0199±.019        | 0.0164±.011        | 0.0233±.005        | 0.0080±.006        |
|                      |             | Diffusion-TS-ICLR2024 | 0.0123±.008        | 0.0122±.009        | 0.0115±.006        | 0.0169±.011        | 0.0193±.017        | 0.0063±.009        | 0.0131±.002        | 0.0711±.019        |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.0509±.004        | 0.0300±.010        | 0.0319±.009        | 0.0349±.009        | 0.0228±.007        | 0.0239±.013        | 0.0231±.000        | 0.0194±.011        |
|                      |             | UPLOTS (Empty)        | 0.0560±.029        | 0.0950±.009        | 0.0889±.008        | 0.0833±.006        | 0.1233±.033        | 0.1210±.074        | 0.0919±.016        | 0.1670±.033        |
|                      |             | UPLOTS (GPT2)         | 0.0066±.004        | <b>0.0099±.004</b> | 0.0077±.004        | 0.0128±.005        | 0.0181±.014        | 0.0059±.009        | 0.0129±.001        | 0.0064±.004        |
|                      |             | UPLOTS (llama3)       | <b>0.0058±.004</b> | 0.0103±.007        | <b>0.0071±.004</b> | <b>0.0116±.004</b> | <b>0.0160±.012</b> | <b>0.0048±.008</b> | <b>0.0121±.000</b> | <b>0.0059±.004</b> |
|                      |             |                       |                    |                    |                    |                    |                    |                    |                    |                    |
| Predictive Score     | Single Data | TimeGAN-NIPS2019      | 0.1607±.000        | 0.1309±.000        | 0.2037±.006        | 0.1900±.004        | 0.1768±.002        | 0.1684±.000        | <b>0.1783±.005</b> | 0.1731±.003        |
|                      |             | TimeVAE-Arxiv2022     | 0.1606±.000        | 0.1309±.002        | 0.2039±.005        | <b>0.1857±.006</b> | 0.1782±.003        | 0.1654±.000        | 0.1785±.001        | 0.1733±.001        |
|                      |             | Diffwave-ICLR2021     | 0.1606±.002        | 0.1310±.001        | 0.2039±.003        | 0.1860±.000        | <b>0.1768±.002</b> | 0.1655±.005        | 0.1784±.003        | 0.1730±.001        |
|                      |             | DiffTime-NIPS2023     | 0.1607±.000        | 0.1310±.000        | 0.2038±.002        | 0.1861±.001        | 0.1784±.001        | 0.1654±.000        | 0.1785±.001        | 0.1733±.000        |
|                      |             | Diffusion-TS-ICLR2024 | 0.1607±.000        | 0.1309±.000        | 0.2037±.004        | 0.1864±.002        | 0.1778±.001        | 0.1654±.001        | 0.1786±.003        | 0.1787±.002        |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.1607±.000        | 0.1310±.000        | 0.2038±.000        | 0.1864±.000        | 0.1777±.000        | 0.1655±.000        | 0.1786±.000        | 0.1735±.000        |
|                      |             | UPLOTS (Empty)        | 0.1614±.001        | 0.1316±.000        | 0.2132±.002        | 0.1960±.002        | 0.1865±.001        | 0.1683±.000        | 0.1873±.001        | 0.1738±.000        |
|                      |             | UPLOTS (GPT2)         | 0.1606±.001        | <b>0.1309±.000</b> | <b>0.2037±.000</b> | 0.1865±.001        | 0.1775±.000        | <b>0.1653±.000</b> | 0.1786±.000        | <b>0.1729±.000</b> |
|                      |             | UPLOTS (llama3)       | <b>0.1605±.000</b> | 0.1310±.000        | <b>0.2037±.001</b> | 0.1864±.000        | 0.1775±.000        | <b>0.1653±.000</b> | 0.1786±.000        | <b>0.1729±.001</b> |
|                      |             |                       |                    |                    |                    |                    |                    |                    |                    |                    |

during the warm-up epochs  $e < 5$ . Let  $\ell_{\max} = \max_i \bar{\ell}_i$  and  $\ell_{\min} = \min_i \bar{\ell}_i$ . Static curriculum weights are then

$$w_i = \begin{cases} s + \frac{\ell_{\max} - \bar{\ell}_i}{\ell_{\max} - \ell_{\min}} (1 - s), & \ell_{\max} > \ell_{\min}, \\ s, & \ell_{\max} = \ell_{\min}. \end{cases} \quad (11)$$

with  $s \in (0, 1)$  denoting the lowest weight. These fixed scores  $w_i^{\text{fixed}}$  down-weight harder datasets early on.

**4.3.2 Rolling-Loss Dynamic Sampling.** Each dataset's deque  $\mathcal{H}_i$  (eq. 10) is reused to derive step-wise sampling probabilities

$$p_i = \frac{\bar{\ell}_i}{\sum_{j=1}^N \bar{\ell}_j}, \quad (12)$$

so batches from higher-loss datasets are drawn more often. A loader is dropped once exhausted, ensuring training continues to target the weakest domains.

The prompt anchors scenario-level constraints, whereas DWTS re-weights batches by sequence difficulty. The two mechanisms are orthogonal yet complementary (see Section 5.3 Ablation Study).

## 5 Experiments

This section presents a comprehensive evaluation of our framework. First, we benchmark state-of-the-art baselines on individual datasets. Next, we assess the base model, Diffusion-TS and our UPLOTS, under multi-dataset training and visualize the fidelity of its generated distributions. Finally, we conduct an ablation study and a

hyper-parameter analysis to validate the robustness and reliability of our approach. All the experimental results are averaged by five times.

**Table 2: Dataset Description**

| Datasets  | Samples | Time Period             | Frequency  |
|-----------|---------|-------------------------|------------|
| ETTh      | 17420   | 01/07/2016 - 26/06/2018 | 10 minutes |
| Energy    | 19711   | 11/01/2016 - 27/05/2016 | 5 minutes  |
| PEMS04    | 19662   | 01/01/2018 - 28/02/2018 | 5 minutes  |
| PEMS08    | 17856   | 01/07/2016 - 31/08/2016 | 5 minutes  |
| ETTh_MP   | 2903    | 01/07/2016 - 26/06/2018 | 1 hour     |
| ETTh_EP   | 2903    | 01/07/2016 - 26/06/2018 | 1 hour     |
| Energy_MP | 2603    | 11/01/2016 - 27/05/2016 | 10 minutes |
| Energy_EP | 2610    | 11/01/2016 - 27/05/2016 | 10 minutes |
| PEMS04_MP | 2183    | 01/01/2018 - 28/02/2018 | 5 minutes  |
| PEMS04_EP | 2183    | 01/01/2018 - 28/02/2018 | 5 minutes  |
| PEMS08_MP | 2294    | 01/07/2016 - 31/08/2016 | 5 minutes  |
| PEMS08_EP | 2294    | 01/07/2016 - 31/08/2016 | 5 minutes  |

### 5.1 Experimental Settings

**Dataset.** We selected four real-world datasets (see Table 2) that provide precise timestamps and thus fit our experiments with constrained conditions. Specifically, we imposed two salient temporal patterns: the morning peak (07:00-10:00) and the evening peak

**Table 3: Comparison with baselines using unified metrics across the full datasets. Lower is better. Best performance is in bold and underlined.**

| Metric               | Data Type | Model                 | ETT                       | Energy                    | PEMS04                    | PEMS08                    |
|----------------------|-----------|-----------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Context-FID          | Single    | TimeGAN-NIPS2019      | 0.1392±.018               | 0.1865±.033               | 0.0492±.008               | 0.0494±.007               |
|                      |           | TimeVAE-Arxiv2022     | 0.6979±.136               | 0.2448±.043               | 0.2400±.041               | 0.1684±.024               |
|                      |           | Diffwave-ICLR2021     | 0.0751±.012               | 0.0814±.008               | 0.0453±.018               | 0.0361±.019               |
|                      |           | DiffTime-NIPS2023     | 0.0576±.017               | 0.0553±.012               | 0.0396±.012               | 0.0371±.021               |
|                      |           | Diffusion-TS-ICLR2024 | 0.0207±.002               | 0.0364±.004               | 0.0418±.004               | 0.0298±.004               |
|                      | Multi     | TimeDP-AAAI2025       | 0.2559±.108               | 0.0702±.007               | 0.0569±.013               | 0.0421±.011               |
|                      |           | UPLOTS (Empty)        | 0.9351±.209               | 0.0807±.009               | 0.1451±.011               | 0.2170±.005               |
|                      |           | UPLOTS (GPT2)         | <b><u>0.0177±.004</u></b> | 0.0358±.006               | <b><u>0.0294±.003</u></b> | <b><u>0.0261±.006</u></b> |
|                      |           | UPLOTS (llama3)       | 0.0181±.005               | <b><u>0.0355±.004</u></b> | 0.0299±.004               | 0.0266±.004               |
|                      |           |                       |                           |                           |                           |                           |
| Discriminative Score | Single    | TimeGAN-NIPS2019      | 0.0241±.022               | 0.0168±.014               | 0.0149±.007               | 0.0197±.010               |
|                      |           | TimeVAE-Arxiv2022     | 0.0273±.018               | 0.0202±.029               | 0.0238±.024               | 0.0180±.011               |
|                      |           | Diffwave-ICLR2021     | 0.0146±.004               | 0.0182±.018               | 0.0161±.002               | 0.0161±.004               |
|                      |           | DiffTime-NIPS2023     | 0.0131±.003               | 0.0153±.011               | 0.0149±.005               | 0.0123±.002               |
|                      |           | Diffusion-TS-ICLR2024 | 0.0168±.008               | 0.0132±.003               | 0.0117±.004               | 0.0096±.001               |
|                      | Multi     | TimeDP-AAAI2025       | 0.0522±.003               | 0.0388±.007               | 0.0145±.011               | 0.0117±.004               |
|                      |           | UPLOTS (Empty)        | 0.1007±.014               | 0.0487±.002               | 0.0544±.026               | 0.0427±.012               |
|                      |           | UPLOTS (GPT2)         | 0.0121±.004               | <b><u>0.0123±.003</u></b> | 0.0120±.005               | 0.0096±.002               |
|                      |           | UPLOTS (llama3)       | <b><u>0.0116±.006</u></b> | 0.0128±.001               | <b><u>0.0112±.002</u></b> | <b><u>0.0092±.003</u></b> |
|                      |           |                       |                           |                           |                           |                           |
| Predictive Score     | Single    | TimeGAN-NIPS2019      | 0.1513±.000               | 0.1876±.000               | 0.2227±.000               | 0.2128±.000               |
|                      |           | TimeVAE-Arxiv2022     | 0.1519±.000               | 0.1877±.000               | 0.2216±.000               | 0.2122±.000               |
|                      |           | Diffwave-ICLR2021     | 0.1515±.000               | 0.1879±.000               | 0.2217±.001               | 0.2123±.000               |
|                      |           | DiffTime-NIPS2023     | 0.1513±.001               | 0.1878±.000               | 0.2215±.000               | 0.2122±.000               |
|                      |           | Diffusion-TS-ICLR2024 | 0.1511±.000               | 0.1878±.000               | 0.2211±.001               | 0.2123±.001               |
|                      | Multi     | TimeDP-AAAI2025       | 0.1511±.001               | 0.1879±.000               | 0.2213±.001               | 0.2123±.000               |
|                      |           | UPLOTS (Empty)        | 0.1576±.001               | 0.1885±.000               | 0.2239±.000               | 0.2122±.000               |
|                      |           | UPLOTS (GPT2)         | <b><u>0.1510±.000</u></b> | <b><u>0.1876±.000</u></b> | 0.2208±.001               | <b><u>0.2120±.000</u></b> |
|                      |           | UPLOTS (llama3)       | <b><u>0.1510±.000</u></b> | <b><u>0.1876±.000</u></b> | <b><u>0.2207±.000</u></b> | <b><u>0.2120±.001</u></b> |
|                      |           |                       |                           |                           |                           |                           |

(17:00-20:00) to ensure our framework can handle the most challenging, high-variance patterns that stress real-world systems. We deliberately omitted sources such as Stock, MuJoCo, and fMRI (utilized by Diffusion-TS [41] and SDformer [4]) due to their less reliable timestamp information. All eight subsets were then employed in Table. 1.

*Metrics.* In our experiments, we quantify synthesis quality along three dimensions: distributional fidelity, temporal feature dependency, and downstream predictive utility using the following evaluation metrics:

- **Discriminative Score** [40] evaluates distributional fidelity by training a binary classifier to differentiate real from generated time series.
- **Predictive Score** [40] measures predictive utility under the Train-on-Synthetic, Test-on-Real (TSTR) paradigm, whereby a sequence forecasting model is trained on synthetic data and evaluated on real observations for next-step accuracy.
- **Context-Fréchet Inception Distance (Context-FID)** [19] assesses temporal and feature dependencies by computing the Fréchet distance between localized subsequence embeddings of real versus synthetic data.

These metrics provide a comprehensive assessment of our generative model’s ability to replicate the statistical properties, inter-variable dynamics, and practical utility of real-world time series.

## 5.2 Main Results

Table 1 reports single-dataset results for the baselines: TimeGAN [40], TimeVAE [7], DiffTime [6], DiffWave [22], Diffusion-TS [41]. From the perspective of multi-dataset training, we selected TimeDP [18] as the primary baseline and compared it with our three UPLOTS variants: (i) UPLOTS (Empty), which is without the Time-Series Prompt Embedding Module (TPERM) and the Dynamic Weighted Training Strategy, (ii) UPLOTS-GPT2, which employs GPT-2 as the frozen LLM, and (iii) UPLOTS-LLaMA, which uses the 1 B-parameter LLaMA 3.2 model. All results are averaged over five runs, and the corresponding variance is reported. The UPLOTS variant without TPERM and the Dynamic Weighted Training Strategy performs poorly, whereas the full UPLOTS achieves stronger results on every dataset and delivers significant metric gains over the basic time series generation models across most metrics. Additionally, more advanced LLM backbones yield even greater improvements.

**Table 4: Ablation study of TPTEM, CALM and RLDS. Lower metrics indicate better performance and the values in bold and underlined indicate the best performance.**

| Dataset                     | UPLOTS               | w/o TPTEM | w/o CALM | w/o RLDS             |
|-----------------------------|----------------------|-----------|----------|----------------------|
| <b>Context-FID</b>          |                      |           |          |                      |
| ETTh_MP                     | <u><b>0.0243</b></u> | 0.1682    | 0.0257   | 0.0247               |
| ETTh_EP                     | <u><b>0.0136</b></u> | 0.1997    | 0.0138   | 0.0156               |
| Energy_MP                   | 0.0348               | 0.2908    | 0.0352   | <u><b>0.0339</b></u> |
| Energy_EP                   | <u><b>0.0334</b></u> | 0.2040    | 0.3350   | 0.0346               |
| PEMS04_MP                   | <u><b>0.0161</b></u> | 0.3285    | 0.0187   | 0.0176               |
| PEMS04_EP                   | <u><b>0.0073</b></u> | 0.6286    | 0.0105   | 0.0080               |
| PEMS08_MP                   | <u><b>0.0137</b></u> | 0.3276    | 0.0156   | 0.0145               |
| PEMS08_EP                   | <u><b>0.0091</b></u> | 0.5256    | 0.1100   | 0.0097               |
| <b>Discriminative Score</b> |                      |           |          |                      |
| ETTh_MP                     | 0.0066               | 0.0560    | 0.0065   | <u><b>0.0061</b></u> |
| ETTh_EP                     | <u><b>0.0099</b></u> | 0.0950    | 0.0102   | 0.0100               |
| Energy_MP                   | <u><b>0.0077</b></u> | 0.0889    | 0.0103   | 0.1090               |
| Energy_EP                   | <u><b>0.0128</b></u> | 0.0833    | 0.0169   | 0.1450               |
| PEMS04_MP                   | <u><b>0.0181</b></u> | 0.1233    | 0.0235   | 0.0210               |
| PEMS04_EP                   | <u><b>0.0059</b></u> | 0.1210    | 0.0076   | 0.0068               |
| PEMS08_MP                   | <u><b>0.0129</b></u> | 0.0919    | 0.0143   | 0.0135               |
| PEMS08_EP                   | <u><b>0.0064</b></u> | 0.1670    | 0.0520   | 0.0069               |

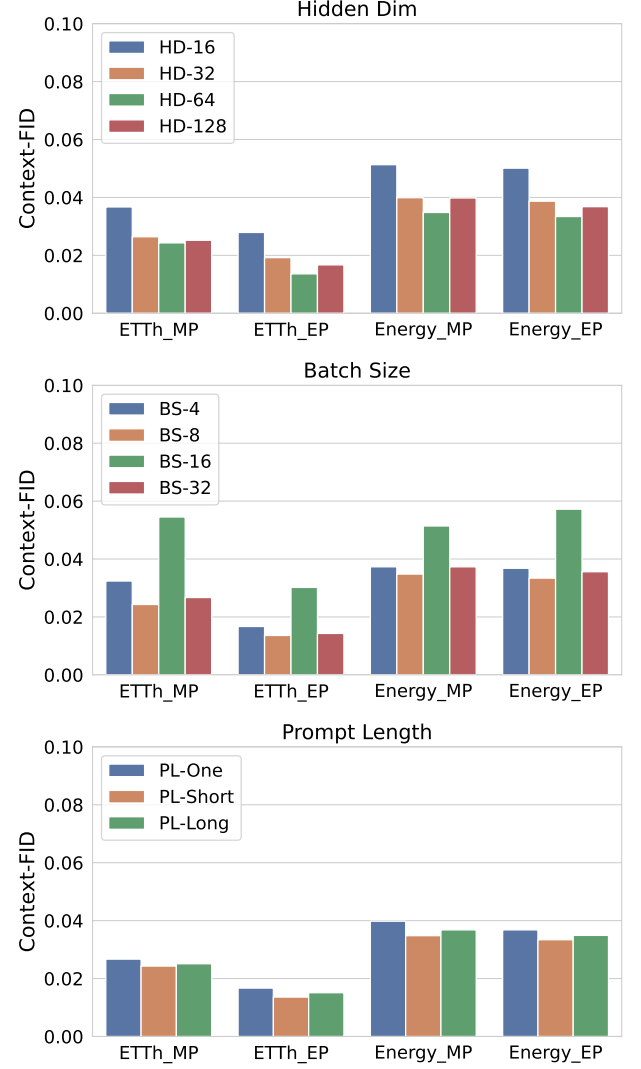
### 5.3 Ablation Study

To demonstrate the individual contributions of our proposed modules, we conduct an ablation study consisting of the full UPLOTS model, UPLOTS without the Time-series Prompt Embedding Module (TPTEM), without the Curriculum-Aware Loss Modulator (CALM), and without Rolling-Loss Dynamic Sampling (RLDS). As shown in Table 4, TPTEM plays a pivotal role, while CALM and RLDS act as useful supplements—helping to balance the various datasets and providing modest performance gains.

### 5.4 Hyper-parameters Analysis

Figure 3 reports the hyper-parameter study, covering hidden dimension, batch size, and TPTEM prompt length. The model under-fits markedly at a hidden dimension of 16 and reaches peak performance at 64, while larger values confer no further benefit. Aside from the smallest batch size of 16, varying batch size has little impact on accuracy. Furthermore, generation quality is largely insensitive to prompt length, indicating that TPTEM remains robust across a broad range of input sizes.

To facilitate reproducibility, we precisely define each level: (1)PL-One: a single-character prompt (e.g., “A”); (2)PL-Short: a concise multi-letter abbreviation (e.g., “ETTHMP”); (3)PL-Long: a complete natural-language phrase (e.g., “The morning peak patterns in the ETTh dataset”). Notably, replacing the full prompt with a fixed token numeric code degrades Context-FID by approximately 5%, confirming that semantic richness not merely length drives the gains. Consequently, we fix hidden dim to be 64 and batch to be 32, and prompt length to be Short for all main results.



**Figure 3: Experimental analysis of hyper-parameters within metric Context-FID.**

### 5.5 Visualization and Analysis

To qualitatively evaluate generative fidelity, we embed both real and synthetic sequences from four representative datasets (*ETTh\_MP/EP* and *Energy\_MP/EP*) into a two-dimensional space using t-SNE, as shown in Figure 4. Across all panels, the point clouds produced by UPLOTS (blue) are tightly interleaved with the ground-truth samples (red), indicating that the model reproduces both the global manifold structure and the local neighbourhood relationships of the original data. The high degree of overlap—especially in the more complex *Energy* datasets—highlights UPLOTS’s ability to capture subtle temporal dependencies that traditional single-dataset generators struggle to model.

We further examine marginal distributions by plotting the probability-density histograms of baselines and our work. As shown in Figure 5, UPLOTS’s density almost perfectly tracks the ground-truth curve

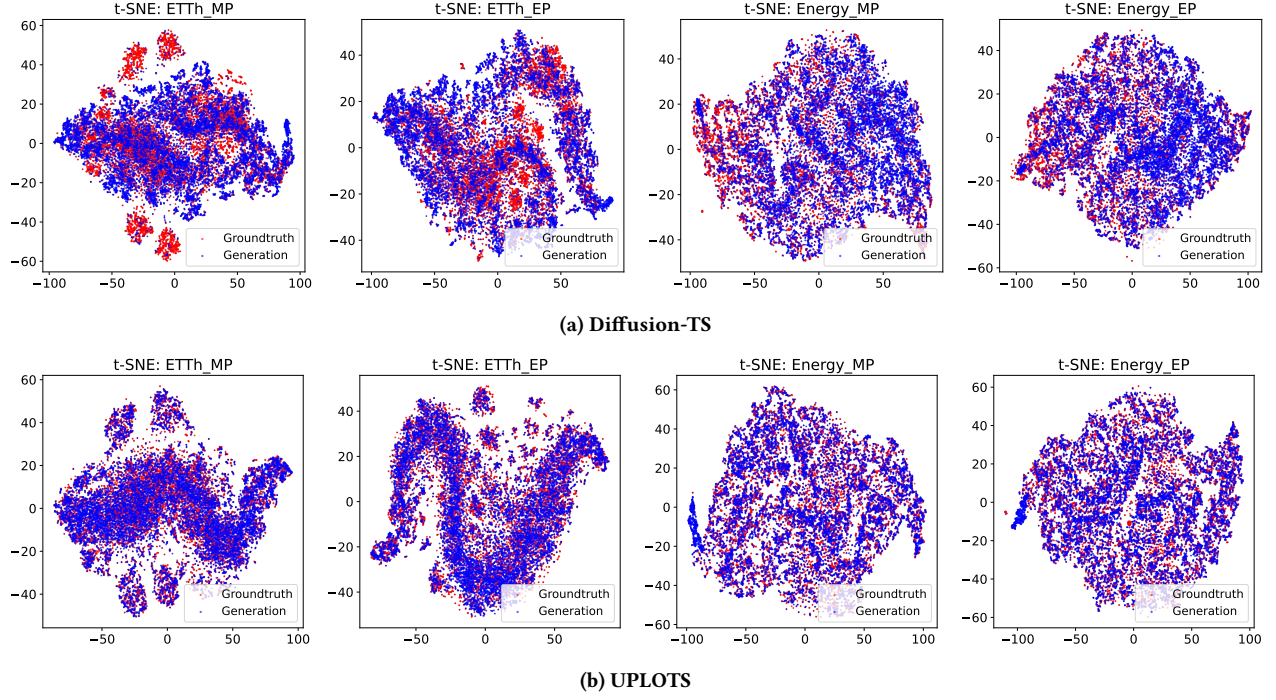


Figure 4: Visualizations of the time series synthesized by (a) Diffusion-TS and (b) UPLOTS.

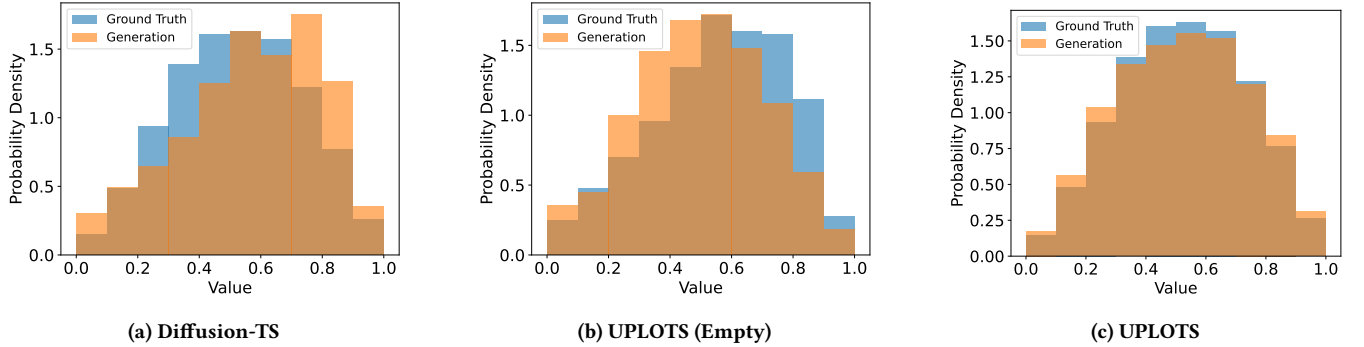


Figure 5: Probability density of basic model, UPLOTS without any techniques and complete UPLOTS.

across the entire value range, underscoring the effectiveness of the proposed techniques in aligning synthetic data with the true distribution.

## 6 Conclusion

This paper introduced UPLOTS, a unified prompt-guided LLM framework for constrained time-series generation that replaces the conventional per-dataset paradigm with a single pretrained backbone augmented by lightweight prompts. By jointly learning heterogeneous temporal structures through multi-dataset training, UPLOTS internalizes domain-specific dynamics and enables controllable generation at inference time without retraining.

Extensive experiments across four benchmarks and two representative peak-pattern scenarios demonstrated that UPLOTS

consistently improved generation fidelity, achieved precise constraint compliance, and exhibited strong cross-domain robustness compared with state-of-the-art and single-dataset models. Notably, these gains were obtained while maintaining training efficiency comparable to standard transformer pipelines.

Future work will explore scaling the framework toward broader temporal domains, richer semantic constraints, and systematic evaluation under true zero-shot distribution shifts, further advancing unified generative modeling for real-world time-series applications.

## References

- [1] Beatrice Acciaio, Stephan Eckstein, and Songyan Hou. 2024. Time-Causal VAE: Robust Financial Time Series Generator. *arXiv preprint arXiv:2411.02947* (2024).
- [2] Yihao Ang, Qiang Huang, Yifan Bao, Anthony KH Tung, and Zhiyong Huang. 2023. TSGBench: Time Series Generation Benchmark. *Proceedings of the VLDB*



- Endowment 17, 3 (2023), 305–318.
- [3] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [4] Zhicheng Chen, FENG SHIBO, Zhong Zhang, Xi Xiao, Xingyu Gao, and Peilin Zhao. 2024. SDformer: Similarity-driven Discrete Transformer For Time Series Generation. *Advances in Neural Information Processing Systems* 37 (2024), 132179–132207.
- [5] Junyoung Chung, Kyle Kastner, Laurent Dinh, Kratarth Goel, Aaron C Courville, and Yoshua Bengio. 2015. A recurrent latent variable model for sequential data. *Advances in neural information processing systems* 28 (2015).
- [6] Andrea Coletta, Sriram Gopalakrishnan, Daniel Borrajo, and Svitlana Vyetrenko. 2023. On the constrained time-series generation problem. *Advances in Neural Information Processing Systems* 36 (2023), 61048–61059.
- [7] Abhyuday Desai, Cynthia Freeman, Zuhui Wang, and Ian Beaver. 2021. Timevae: A variational auto-encoder for multivariate time series generation. *arXiv preprint arXiv:2111.08095* (2021).
- [8] Cristóbal Esteban, Stephanie L Hyland, and Gunnar Rätsch. 2017. Real-valued (medical) time series generation with recurrent conditional gans. *arXiv preprint arXiv:1706.02633* (2017).
- [9] Otto Fabius and Joost R Van Amersfoort. 2014. Variational recurrent auto-encoders. *arXiv preprint arXiv:1412.6581* (2014).
- [10] Vincent Fortuin, Dmitry Baranchuk, Gunnar Raetsch, and Stephan Mandt. 2020. GP-VAE: Deep Probabilistic Time Series Imputation. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 108)*, Silvia Chiappa and Roberto Calandra (Eds.). PMLR, 1651–1661. <https://proceedings.mlr.press/v108/fortuin20a.html>
- [11] Marco Fraccaro, Søren Kaae Sønderby, Ulrich Paquet, and Ole Winther. 2016. Sequential neural models with stochastic layers. *Advances in neural information processing systems* 29 (2016).
- [12] Asadullah Hill Galib, Pang-Ning Tan, and Lifeng Luo. 2024. FIDE: Frequency-Inflated Conditional Diffusion Model for Extreme-Aware Time Series Generation. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.
- [13] Hongfan Gao, Wangmeng Shen, Xiangfei Qiu, Ronghui Xu, Jilin Hu, and Bin Yang. 2024. Diffmp: Efficient diffusion model for probabilistic time series imputation with bidirectional mamba backbone. *arXiv preprint arXiv:2410.13338* (2024).
- [14] Azul Garza, Cristian Challu, and Max Mergenthaler-Canseco. 2023. TimeGPT-1. *arXiv preprint arXiv:2310.03589* (2023).
- [15] Yunfeng Ge, Jiawei Li, Yiji Zhao, Haomin Wen, Zhao Li, Meikang Qiu, Hongyan Li, Ming Jin, and Shirui Pan. 2025. T2S: High-resolution Time Series Generation with Text-to-Series Diffusion Models. *arXiv preprint arXiv:2505.02417* (2025).
- [16] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).
- [17] Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew G Wilson. 2023. Large language models are zero-shot time series forecasters. *Advances in Neural Information Processing Systems* 36 (2023), 19622–19635.
- [18] Yu-Hao Huang, Chang Xu, Yueying Wu, Wu-Jun Li, and Jiang Bian. 2025. Timedp: Learning to generate multi-domain time series with domain prompts. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 17520–17527.
- [19] Paul Jeha, Michael Bohlke-Schneider, Pedro Mercado, Shubham Kapoor, Rajbir Singh Nirwan, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. 2022. PSA-GAN: Progressive Self Attention GANs for Synthetic Time Series. In *International Conference on Learning Representations*.
- [20] Jinsung Jeon, Jeonghak Kim, Haryong Song, Seunghyeon Cho, and Noseong Park. 2022. GT-GAN: General purpose time series synthesis with generative adversarial networks. *Advances in Neural Information Processing Systems* 35 (2022), 36999–37010.
- [21] Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-fang Li, Shirui Pan, et al. 2024. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models. In *International Conference on Learning Representations*.
- [22] Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. 2021. DiffWave: A Versatile Diffusion Model for Audio Synthesis. In *International Conference on Learning Representations*.
- [23] Daesoo Lee, Sara Malacarne, and Erlend Aune. 2023. Vector Quantized Time Series Generation with a Bidirectional Prior Model. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 7665–7693.
- [24] Hao Li, Yu-Hao Huang, Chang Xu, Viktor Schlegel, Ren-He Jiang, Riza Batista-Navarro, Goran Nenadic, and Jiang Bian. 2025. Bridge: Bootstrapping text to control time-series generation via multi-agent iterative optimization and diffusion modelling. *arXiv preprint arXiv:2503.02445* (2025).
- [25] Yang Li, Han Meng, Zhenyu Bi, Ingoly T. Urnes, and Haipeng Chen. 2025. Population Aware Diffusion for Time Series Generation. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 17 (Apr. 2025), 18520–18529. doi:10.1609/aaai.v39i17.34038
- [26] Xu Liu, Junfeng Hu, Yuan Li, Shizhe Diao, Yuxuan Liang, Bryan Hooi, and Roger Zimmermann. 2024. Unitime: A language-empowered unified model for cross-domain time series forecasting. In *Proceedings of the ACM Web Conference 2024*. 4095–4106.
- [27] Olof Mogren. 2016. C-RNN-GAN: Continuous recurrent neural networks with adversarial training. *arXiv preprint arXiv:1611.09904* (2016).
- [28] Abdulmajid Murad and Massimiliano Ruocco. 2025. Synthetic Aircraft Trajectory Generation Using Time-Based VQ-VAE. *arXiv preprint arXiv:2504.09101* (2025).
- [29] Ilan Naiman, Nimrod Berman, Itai Pemper, Idan Arbiv, Gal Fadlon, and Omri Azencot. 2024. Utilizing image transforms and diffusion models for generative modeling of short and long time series. *Advances in Neural Information Processing Systems* 37 (2024), 121699–121730.
- [30] Ilan Naiman, N Benjamin Erichson, Pu Ren, Michael W Mahoney, and Omri Azencot. 2024. GENERATIVE MODELING OF REGULAR AND IRREGULAR TIME SERIES DATA VIA KOOPMAN VAES. In *12th International Conference on Learning Representations, ICLR 2024*.
- [31] Chao Pang, Xinzhuo Jiang, Nishanth Parameshwar Pavinkurve, Krishna S Kalluri, Elise L Minto, Jason Patterson, Linying Zhang, George Hripsak, Gamze Gürsoy, Noémie Elhadad, et al. 2024. CEHR-GPT: Generating electronic health records with chronological patient timelines. *arXiv preprint arXiv:2402.04400* (2024).
- [32] Hengzhi Pei, Kan Ren, Yuqing Yang, Chang Liu, Tao Qin, and Dongsheng Li. 2021. Towards generating real-world time series data. In *2021 IEEE International Conference on Data Mining (ICDM)*. IEEE, 469–478.
- [33] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. *OpenAI Blog* 1, 8 (2019). [https://cdn.openai.com/better-language-models/language\\_models\\_are\\_unsupervised\\_multitask\\_learners.pdf](https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf)
- [34] Aditya Shankar, Lydia Y Chen, Arie van Deursen, and Rihan Hai. 2025. WaveStitch: Flexible and Fast Conditional Time Series Generation with Diffusion Models. *arXiv preprint arXiv:2503.06231* (2025).
- [35] Namjoon Suh, Yuning Yang, Din-Yin Hsieh, Qitong Luan, Shirong Xu, Shixiang Zhu, and Guang Cheng. 2024. TimeAutoDiff: Combining Autoencoder and Diffusion model for time series tabular data synthesizing. *arXiv preprint arXiv:2406.16028* (2024).
- [36] Chenxi Sun, Hongyan Li, Yaliang Li, and Shenda Hong. 2024. TEST: Text Prototype Aligned Embedding to Activate LLM’s Ability for Time Series. In *The Twelfth International Conference on Learning Representations*.
- [37] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmin Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shriti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288* (2023).
- [38] Tianlin Xu, Li Kevin Wenliang, Michael Munn, and Beatrice Acciaio. 2020. Cotgan: Generating sequential data via causal optimal transport. *Advances in neural information processing systems* 33 (2020), 8798–8809.
- [39] Tijin Yan, Hengheng Gong, He Yongping, Yufeng Zhan, and Yuanqing Xia. 2024. Probabilistic time series modeling with decomposable denoising diffusion model. In *Forty-first International Conference on Machine Learning*.
- [40] Jinsung Yoon, Daniel Jarrett, and Mihaela Van der Schaar. 2019. Time-series generative adversarial networks. *Advances in neural information processing systems* 32 (2019).
- [41] Xinyu Yuan and Yan Qiao. 2024. Diffusion-TS: Interpretable Diffusion for General Time Series Generation. In *The Twelfth International Conference on Learning Representations*.
- [42] Tian Zhou, Peisong Niu, Liang Sun, Rong Jin, et al. 2023. One fits all: Power general time series analysis by pretrained lm. *Advances in neural information processing systems* 36 (2023), 43322–43355.

## A Appendix

### A.1 Results

To complement the main results, the Appendix reports four additional quantitative comparisons under alternative generation lengths and evaluation protocols.

Specifically, tables 5 and 6 present the baseline comparisons under generation lengths of 48 and 96, respectively.

In addition, tables 7 and 8 summarize the unified-metric evaluation on the full datasets under the same two lengths (48 and 96).

Overall, these supplementary tables further validate that the observations and relative trends discussed in the main paper remained consistent across sequence lengths and evaluation settings.

Table 5: Comparison with baselines under the setting of generation length 48.

| Metric               | Data Type   | Model                 | ETTh_MP             | ETTh_EP             | Energy_MP           | Energy_EP           | PEMS04_MP           | PEMS04_EP           | PEMS08_MP           | PEMS08_EP           |
|----------------------|-------------|-----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Context-FID          | Single Data | TimeGAN-NIPS2019      | 0.9031±.093         | 0.3231±.066         | 0.8046±.122         | 0.5361±.037         | 0.0990±.011         | 0.1058±.017         | 0.3113±.038         | 0.4196±.039         |
|                      |             | TimeVAE-Arxiv2022     | 0.8859±.072         | 0.5697±.067         | 0.5770±.061         | 0.6614±.073         | 0.7822±.124         | 0.7798±.086         | 0.8289±.126         | 0.6162±.059         |
|                      |             | Diffwave-ICLR2021     | 0.2156±.047         | 0.1466±.012         | 0.0969±.009         | 0.0975±.013         | 0.0941±.006         | 0.0996±.007         | 0.0845±.012         | 0.1005±.016         |
|                      |             | DiffTime-NIPS2023     | 0.2058±.045         | 0.1214±.009         | 0.0839±.013         | 0.0872±.009         | 0.0814±.004         | 0.0895±.010         | 0.0741±.005         | 0.0942±.012         |
|                      |             | Diffusion-TS-ICLR2024 | 0.2042±.044         | 0.2123±.035         | 0.0825±.015         | 0.0760±.007         | 0.0491±.002         | 0.0221±.004         | 0.0389±.003         | 0.0212±.001         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.7200±.197         | 0.1003±.060         | 0.0369±.005         | 0.0383±.006         | 0.3669±.099         | 0.3246±.075         | 0.2694±.047         | 0.1547±.030         |
|                      |             | UPLOTS (Empty)        | 0.2083±.063         | 0.2599±.046         | 0.3331±.041         | 0.2445±.024         | 0.8989±.058         | 1.5606±.155         | 0.7115±.142         | 1.2696±.085         |
|                      |             | UPLOTS (GPT2)         | 0.0878±.009         | <b>0.0136</b> ±.001 | 0.0348±.004         | 0.0334±.002         | 0.0188±.003         | <b>0.0121</b> ±.002 | <b>0.0152</b> ±.001 | <b>0.0122</b> ±.003 |
|                      |             | UPLOTS (llama3)       | <b>0.0854</b> ±.003 | 0.0145±.002         | <b>0.0323</b> ±.002 | <b>0.0315</b> ±.002 | <b>0.0167</b> ±.003 | 0.0134±.005         | 0.0177±.003         | 0.0143±.002         |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |
| Discriminative Score | Single Data | TimeGAN-NIPS2019      | 0.0492±.059         | 0.0215±.022         | 0.0534±.039         | 0.0126±.006         | 0.0052±.003         | 0.0144±.017         | 0.0319±.025         | 0.0325±.023         |
|                      |             | TimeVAE-Arxiv2022     | 0.0462±.039         | 0.0192±.019         | 0.0361±.030         | 0.0440±.029         | 0.0393±.025         | 0.0207±.026         | 0.0289±.024         | 0.0495±.051         |
|                      |             | Diffwave-ICLR2021     | 0.0426±.016         | 0.0194±.021         | 0.0199±.003         | 0.0293±.018         | 0.0224±.015         | 0.0184±.024         | 0.0214±.016         | 0.0201±.021         |
|                      |             | DiffTime-NIPS2023     | 0.0395±.015         | 0.0179±.016         | 0.0147±.004         | 0.0273±.019         | 0.0182±.013         | 0.0139±.022         | 0.0191±.013         | 0.0169±.018         |
|                      |             | Diffusion-TS-ICLR2024 | 0.0941±.115         | 0.0890±.030         | 0.0373±.017         | 0.0430±.017         | 0.0374±.003         | 0.0221±.010         | 0.0229±.005         | 0.0233±.030         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.0389±.034         | 0.0324±.028         | 0.0124±.008         | 0.0297±.025         | 0.0193±.012         | 0.0296±.025         | 0.0177±.012         | 0.0201±.008         |
|                      |             | UPLOTS (Empty)        | 0.1105±.074         | 0.1012±.014         | 0.0921±.008         | 0.0892±.002         | 0.1563±.068         | 0.2103±.091         | 0.1749±.017         | 0.2040±.036         |
|                      |             | UPLOTS (GPT2)         | 0.0236±.003         | <b>0.0136</b> ±.005 | 0.0127±.005         | 0.0238±.004         | 0.0203±.016         | <b>0.0129</b> ±.007 | 0.0113±.003         | 0.0125±.003         |
|                      |             | UPLOTS (llama3)       | <b>0.0227</b> ±.005 | 0.0143±.005         | <b>0.0116</b> ±.003 | <b>0.0229</b> ±.003 | <b>0.0184</b> ±.011 | 0.0133±.005         | <b>0.0111</b> ±.002 | <b>0.0116</b> ±.001 |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |
| Predictive Score     | Single Data | TimeGAN-NIPS2019      | 0.1727±.002         | 0.1308±.000         | 0.2162±.001         | 0.1858±.000         | <b>0.1759</b> ±.000 | 0.1659±.000         | 0.1800±.000         | 0.1734±.000         |
|                      |             | TimeVAE-Arxiv2022     | 0.1615±.000         | 0.1307±.000         | 0.2037±.000         | <b>0.1853</b> ±.000 | 0.1776±.000         | 0.1652±.000         | 0.1783±.000         | 0.1735±.000         |
|                      |             | Diffwave-ICLR2021     | 0.1617±.000         | 0.1310±.001         | 0.2036±.000         | 0.1865±.001         | 0.1757±.000         | 0.1652±.000         | 0.1788±.000         | 0.1732±.002         |
|                      |             | DiffTime-NIPS2023     | 0.1617±.000         | 0.1309±.000         | 0.2034±.000         | 0.1863±.000         | 0.1757±.000         | 0.1651±.000         | 0.1785±.000         | 0.1731±.000         |
|                      |             | Diffusion-TS-ICLR2024 | 0.1609±.000         | 0.1309±.000         | 0.2034±.000         | 0.1861±.000         | 0.1773±.001         | 0.1652±.000         | 0.1788±.000         | 0.1730±.000         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.1616±.000         | 0.1309±.000         | 0.2035±.000         | 0.1865±.000         | 0.1775±.000         | 0.1653±.000         | 0.1784±.000         | 0.1731±.000         |
|                      |             | UPLOTS (Empty)        | 0.1613±.000         | 0.1312±.000         | 0.2135±.001         | 0.1967±.001         | 0.1855±.001         | 0.1679±.001         | 0.1874±.000         | 0.1740±.000         |
|                      |             | UPLOTS (GPT2)         | 0.1615±.001         | <b>0.1307</b> ±.000 | <b>0.2033</b> ±.000 | 0.1865±.001         | 0.1775±.000         | 0.1651±.000         | <b>0.1782</b> ±.000 | <b>0.1727</b> ±.000 |
|                      |             | UPLOTS (llama3)       | <b>0.1614</b> ±.000 | 0.1308±.000         | <b>0.2033</b> ±.001 | 0.1864±.000         | 0.1775±.000         | <b>0.1650</b> ±.000 | <b>0.1782</b> ±.000 | <b>0.1727</b> ±.001 |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |

Table 6: Comparison with baselines under the setting of generation length 96.

| Metric               | Data Type   | Model                 | ETTh_MP             | ETTh_EP             | Energy_MP           | Energy_EP           | PEMS04_MP           | PEMS04_EP           | PEMS08_MP           | PEMS08_EP           |
|----------------------|-------------|-----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Context-FID          | Single Data | TimeGAN-NIPS2019      | 1.3651±.174         | 2.5919±.332         | 1.4668±.156         | 2.3660±.428         | 1.2871±.083         | 0.3040±.035         | 1.9413±.231         | 2.5267±.212         |
|                      |             | TimeVAE-Arxiv2022     | 0.9402±.228         | 0.7119±.119         | 0.7662±.072         | 0.8743±.067         | 0.9719±.157         | 0.7564±.072         | 0.8939±.093         | 0.4741±.031         |
|                      |             | Diffwave-ICLR2021     | 0.2954±.075         | 0.1642±.028         | 0.1512±.019         | 0.1764±.035         | 0.1241±.012         | 0.1314±.011         | 0.1025±.015         | 0.1120±.008         |
|                      |             | DiffTime-NIPS2023     | 0.2637±.067         | 0.1560±.021         | 0.1475±.017         | 0.1521±.022         | 0.1020±.009         | 0.1208±.009         | 0.0942±.011         | 0.0927±.010         |
|                      |             | Diffusion-TS-ICLR2024 | 0.2493±.066         | 0.3479±.084         | 0.1452±.016         | 0.1312±.007         | 0.1016±.015         | <b>0.0376</b> ±.005 | 0.1010±.012         | 0.0506±.004         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.4402±.082         | 0.2665±.062         | 0.1962±.056         | 0.2039±.039         | 0.7131±.114         | 0.5951±.096         | 0.5049±.069         | 0.3950±.120         |
|                      |             | UPLOTS (Empty)        | 0.2283±.049         | 0.3537±.073         | 0.4295±.050         | 0.3270±.017         | 1.8241±.118         | 2.7541±.394         | 1.2804±.150         | 2.2246±.160         |
|                      |             | UPLOTS (GPT2)         | <b>0.1335</b> ±.013 | <b>0.0896</b> ±.018 | 0.1443±.025         | 0.1316±.020         | 0.0423±.011         | 0.0725±.013         | <b>0.0436</b> ±.014 | <b>0.0406</b> ±.011 |
|                      |             | UPLOTS (llama3)       | 0.1346±.012         | 0.0900±.008         | <b>0.1432</b> ±.022 | <b>0.1311</b> ±.015 | <b>0.0412</b> ±.009 | 0.0712±.021         | 0.0446±.016         | 0.0413±.015         |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |
| Discriminative Score | Single Data | TimeGAN-NIPS2019      | 0.0839±.061         | 0.2331±.043         | 0.0437±.043         | 0.0926±.034         | 0.0393±.030         | 0.0252±.020         | 0.0457±.095         | 0.1023±.030         |
|                      |             | TimeVAE-Arxiv2022     | 0.0248±.033         | 0.0154±.016         | 0.0328±.033         | 0.0346±.034         | 0.0577±.035         | 0.0438±.049         | 0.0346±.009         | 0.0329±.028         |
|                      |             | Diffwave-ICLR2021     | 0.0264±.027         | 0.0168±.015         | 0.0154±.021         | 0.0201±.016         | 0.0196±.011         | 0.0145±.005         | 0.0175±.005         | 0.0184±.020         |
|                      |             | DiffTime-NIPS2023     | 0.0235±.024         | 0.0136±.013         | 0.0126±.022         | 0.0160±.012         | 0.0183±.007         | 0.0123±.003         | 0.0139±.004         | 0.0147±.019         |
|                      |             | Diffusion-TS-ICLR2024 | 0.2075±.049         | 0.0698±.018         | 0.0438±.017         | 0.0375±.035         | 0.0421±.007         | 0.0150±.007         | 0.0327±.022         | 0.0136±.023         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.0689±.008         | 0.0512±.012         | 0.0599±.006         | 0.0608±.002         | 0.0753±.015         | 0.0239±.005         | 0.0264±.011         | 0.0195±.008         |
|                      |             | UPLOTS (Empty)        | 0.0947±.078         | 0.1030±.035         | 0.1021±.005         | 0.1016±.003         | 0.1865±.085         | 0.2283±.097         | 0.1646±.071         | 0.1661±.083         |
|                      |             | UPLOTS (GPT2)         | 0.0224±.002         | <b>0.0111</b> ±.003 | 0.0106±.005         | 0.0136±.003         | 0.0101±.004         | 0.0091±.004         | 0.0136±.004         | 0.0121±.001         |
|                      |             | UPLOTS (llama3)       | <b>0.0197</b> ±.005 | 0.0114±.006         | <b>0.0105</b> ±.003 | <b>0.0116</b> ±.004 | <b>0.0092</b> ±.002 | <b>0.0085</b> ±.003 | <b>0.0131</b> ±.001 | <b>0.0116</b> ±.002 |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |
| Predictive Score     | Single Data | TimeGAN-NIPS2019      | 0.1744±.001         | 0.1348±.000         | 0.2131±.002         | 0.2142±.002         | 0.1788±.001         | 0.1656±.001         | 0.1802±.001         | 0.1782±.001zhen     |
|                      |             | TimeVAE-Arxiv2022     | 0.1603±.000         | 0.1303±.000         | 0.2033±.000         | <b>0.1847</b> ±.000 | 0.1781±.000         | 0.1650±.000         | 0.1786±.000         | 0.1740±.000         |
|                      |             | Diffwave-ICLR2021     | 0.1608±.000         | 0.1304±.000         | 0.2031±.000         | 0.1858±.001         | 0.1759±.000         | 0.1647±.000         | 0.1786±.000         | 0.1732±.002         |
|                      |             | DiffTime-NIPS2023     | 0.1607±.000         | 0.1304±.000         | 0.2030±.000         | 0.1858±.000         | <b>0.1757</b> ±.000 | 0.1646±.000         | 0.1786±.000         | 0.1731±.000         |
|                      |             | Diffusion-TS-ICLR2024 | 0.1607±.000         | 0.1317±.000         | 0.2033±.000         | 0.1857±.000         | 0.1776±.000         | 0.1648±.000         | 0.1797±.000         | 0.1731±.000         |
|                      | Multi Data  | TimeDP-AAAI2025       | 0.1606±.000         | 0.1309±.000         | 0.2032±.001         | 0.1858±.001         | 0.1776±.000         | 0.1647±.000         | 0.1789±.000         | 0.1736±.001         |
|                      |             | UPLOTS (Empty)        | 0.1611±.000         | 0.1310±.000         | 0.2138±.000         | 0.1958±.001         | 0.1854±.001         | 0.1678±.000         | 0.1878±.001         | 0.1742±.000         |
|                      |             | UPLOTS (GPT2)         | <b>0.1603</b> ±.001 | <b>0.1303</b> ±.000 | <b>0.2030</b> ±.000 | 0.1859±.001         | 0.1775±.000         | <b>0.1645</b> ±.000 | <b>0.1783</b> ±.000 | <b>0.1728</b> ±.000 |
|                      |             | UPLOTS (llama3)       | <b>0.1603</b> ±.000 | 0.1304±.000         | 0.2031±.001         | 0.1859±.000         | 0.1775±.000         | <b>0.1645</b> ±.000 | <b>0.1783</b> ±.000 | <b>0.1728</b> ±.001 |
|                      |             |                       |                     |                     |                     |                     |                     |                     |                     |                     |

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

**Table 7: Comparison with baselines using unified metrics across the full datasets under the length of 48.**

| Metric               | Data Type | Model                 | ETT                | Energy             | PEMS04             | PEMS08             |
|----------------------|-----------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Context-FID          | Single    | TimeGAN-NIPS2019      | 0.4350±.054        | 0.2574±.049        | 0.0852±.016        | 0.1194±.024        |
|                      |           | TimeVAE-Arxiv2022     | 0.5465±.048        | 0.4577±.045        | 0.1128±.009        | 0.0811±.016        |
|                      |           | Diffwave-ICLR2021     | 0.1562±.021        | 0.0672±.009        | 0.1196±.009        | 0.0472±.007        |
|                      |           | DiffTime-NIPS2023     | 0.1345±.014        | 0.0616±.006        | 0.1128±.009        | 0.0421±.004        |
|                      |           | Diffusion-TS-ICLR2024 | 0.0811±.014        | 0.0597±.008        | 0.0586±.014        | 0.9696±.146        |
|                      | Multi     | TimeDP-AAAI2025       | 0.3684±.109        | 0.0849±.089        | 0.0947±.091        | 0.0547±.064        |
|                      |           | UPLOTS (Empty)        | 0.9351±.209        | 0.0807±.009        | 0.1236±.015        | 0.1125±.003        |
|                      |           | UPLOTS (GPT2)         | 0.0810±.002        | 0.0585±.006        | 0.0574±.008        | 0.0328±.005        |
|                      |           | UPLOTS (llama3)       | <b>0.0802±.001</b> | <b>0.0579±.005</b> | <b>0.0410±.004</b> | <b>0.0296±.004</b> |
|                      |           |                       |                    |                    |                    |                    |
| Discriminative Score | Single    | TimeGAN-NIPS2019      | 0.0307±.009        | 0.0165±.012        | 0.0121±.002        | 0.0359±.007        |
|                      |           | TimeVAE-Arxiv2022     | 0.0238±.035        | 0.0143±.017        | 0.0129±.005        | 0.0113±.001        |
|                      |           | Diffwave-ICLR2021     | 0.0145±.003        | 0.0176±.009        | 0.0157±.003        | 0.0148±.005        |
|                      |           | DiffTime-NIPS2023     | 0.0131±.003        | 0.0153±.011        | 0.0149±.005        | 0.0123±.002        |
|                      |           | Diffusion-TS-ICLR2024 | 0.0138±.007        | 0.0174±.002        | 0.0133±.005        | 0.0142±.006        |
|                      | Multi     | TimeDP-AAAI2025       | 0.0516±.011        | 0.0396±.010        | 0.0179±.021        | 0.0144±.005        |
|                      |           | UPLOTS (Empty)        | 0.1007±.014        | 0.0670±.002        | 0.0640±.026        | 0.0127±.012        |
|                      |           | UPLOTS (GPT2)         | 0.0121±.003        | 0.0133±.004        | 0.0120±.001        | <b>0.0109±.003</b> |
|                      |           | UPLOTS (llama3)       | <b>0.0119±.006</b> | <b>0.0128±.001</b> | <b>0.0119±.002</b> | 0.0111±.004        |
|                      |           |                       |                    |                    |                    |                    |
| Predictive Score     | Single    | TimeGAN-NIPS2019      | 0.1513±.000        | 0.1877±.000        | 0.2228±.001        | 0.2149±.000        |
|                      |           | TimeVAE-Arxiv2022     | 0.1511±.000        | <b>0.1875±.000</b> | 0.2210±.000        | 0.2121±.000        |
|                      |           | Diffwave-ICLR2021     | 0.1513±.000        | 0.1876±.000        | 0.2216±.001        | 0.2123±.000        |
|                      |           | DiffTime-NIPS2023     | 0.1512±.001        | 0.1876±.000        | 0.2215±.000        | 0.2122±.000        |
|                      |           | Diffusion-TS-ICLR2024 | 0.1511±.000        | 0.1877±.000        | 0.2214±.001        | 0.2121±.001        |
|                      | Multi     | TimeDP-AAAI2025       | 0.1513±.000        | 0.1877±.001        | 0.2215±.000        | 0.2122±.000        |
|                      |           | UPLOTS (Empty)        | 0.1576±.001        | 0.1885±.000        | 0.2239±.000        | 0.2122±.000        |
|                      |           | UPLOTS (GPT2)         | 0.1510±.000        | 0.1876±.000        | 0.2209±.001        | 0.2120±.000        |
|                      |           | UPLOTS (llama3)       | <b>0.1510±.000</b> | 0.1876±.000        | <b>0.2208±.000</b> | <b>0.2119±.001</b> |
|                      |           |                       |                    |                    |                    |                    |

**Table 8: Comparison with baselines using unified metrics across the full datasets under the length of 96.**

| Metric               | Data Type | Model                 | ETT                | Energy             | PEMS04             | PEMS08             |
|----------------------|-----------|-----------------------|--------------------|--------------------|--------------------|--------------------|
| Context-FID          | Single    | TimeGAN-NIPS2019      | 2.4917±.523        | 0.4289±.052        | 4.7468±.644        | 0.1685±.028        |
|                      |           | TimeVAE-Arxiv2022     | 0.9294±.102        | 0.3494±.050        | 0.1732±.024        | 0.1483±.014        |
|                      |           | Diffwave-ICLR2021     | 0.1963±.031        | 0.0615±.007        | 0.1841±.035        | 0.1684±.042        |
|                      |           | DiffTime-NIPS2023     | 0.1855±.033        | 0.0561±.008        | 0.1654±.025        | 0.1574±.031        |
|                      |           | Diffusion-TS-ICLR2024 | 0.2285±.054        | 0.0804±.007        | 0.0431±.006        | 0.0395±.005        |
|                      | Multi     | TimeDP-AAAI2025       | 0.2325±.121        | 0.0904±.029        | 0.0684±.042        | 0.0541±.023        |
|                      |           | UPLOTS (Empty)        | 0.9351±.209        | 0.0807±.029        | 0.1451±.051        | 0.2177±.035        |
|                      |           | UPLOTS (GPT2)         | 0.0394±.008        | 0.0358±.006        | <b>0.0399±.003</b> | <b>0.0291±.006</b> |
|                      |           | UPLOTS (llama3)       | <b>0.0387±.004</b> | <b>0.0355±.004</b> | 0.0410±.004        | 0.0296±.004        |
|                      |           |                       |                    |                    |                    |                    |
| Discriminative Score | Single    | TimeGAN-NIPS2019      | 0.0129±.026        | 0.0260±.013        | 0.2639±.108        | 0.0226±.003        |
|                      |           | TimeVAE-Arxiv2022     | 0.0133±.008        | 0.0241±.022        | 0.0116±.007        | 0.0148±.013        |
|                      |           | Diffwave-ICLR2021     | 0.0156±.004        | 0.0152±.002        | 0.0134±.003        | 0.0164±.003        |
|                      |           | DiffTime-NIPS2023     | 0.0133±.005        | 0.0119±.003        | 0.0115±.004        | 0.0149±.004        |
|                      |           | Diffusion-TS-ICLR2024 | 0.0670±.020        | 0.0301±.015        | 0.0292±.014        | 0.0297±.019        |
|                      | Multi     | TimeDP-AAAI2025       | 0.0596±.005        | 0.0298±.007        | 0.0218±.006        | 0.0224±.003        |
|                      |           | UPLOTS (Empty)        | 0.1445±.029        | 0.0853±.018        | 0.0746±.031        | 0.0469±.022        |
|                      |           | UPLOTS (GPT2)         | 0.0125±.003        | 0.0113±.001        | 0.0110±.003        | 0.0134±.002        |
|                      |           | UPLOTS (llama3)       | <b>0.0124±.002</b> | <b>0.0112±.001</b> | <b>0.0108±.002</b> | <b>0.0124±.003</b> |
|                      |           |                       |                    |                    |                    |                    |
| Predictive Score     | Single    | TimeGAN-NIPS2019      | 0.1513±.000        | 0.1876±.000        | 0.2246±.000        | 0.2125±.000        |
|                      |           | TimeVAE-Arxiv2022     | 0.1511±.000        | <b>0.1873±.000</b> | 0.2214±.000        | 0.2122±.000        |
|                      |           | Diffwave-ICLR2021     | 0.1512±.000        | 0.1879±.000        | 0.2214±.000        | 0.2123±.001        |
|                      |           | DiffTime-NIPS2023     | 0.1513±.000        | 0.1877±.000        | 0.2212±.000        | 0.2122±.000        |
|                      |           | Diffusion-TS-ICLR2024 | 0.1511±.000        | 0.1876±.000        | 0.2213±.001        | 0.2122±.000        |
|                      | Multi     | TimeDP-AAAI2025       | 0.1512±.000        | 0.1877±.001        | 0.2214±.000        | 0.2122±.000        |
|                      |           | UPLOTS (Empty)        | 0.1567±.001        | 0.1885±.000        | 0.2238±.000        | 0.2121±.000        |
|                      |           | UPLOTS (GPT2)         | 0.1510±.000        | 0.1876±.000        | <b>0.2211±.000</b> | 0.2121±.000        |
|                      |           | UPLOTS (llama3)       | <b>0.1510±.000</b> | 0.1876±.000        | <b>0.2211±.000</b> | <b>0.2120±.000</b> |
|                      |           |                       |                    |                    |                    |                    |